Mathematical Architecture of the Twisted Partner Ring, Discrete Logic Routing, and Unitary Compilation in Biological Landscapes

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Theoretical Foundation: The Single Inverter and the Nested Partner Ring

The operational framework under evaluation interfaces two mathematical methodologies: an exact, finite-combinatorial logic router grounded in twisted register transformations and involution rings, and a unitary quantum simulation model designed to compile conformational state transitions across molecular energy landscapes. Decoupling the verified mechanics of this system from its broader theoretical claims requires analyzing the substrate from its axiomatic foundations.

The Forced Loop and the Principle of Non-Terminal Continuation

The classical combinatorial layer originates from an operational constraint on state machines designated as continuation: no permissible computational state may be terminal, and every operation must preserve an exact return path without recourse to external tracking memory. On a single-bit cell, exactly four state transformations exist: the identity map, two constant reset functions ($0 \mapsto 0, 1 \mapsto 0$ and $0 \mapsto 1, 1 \mapsto 1$), and the logical complement (NOT). The identity and constant maps possess fixed states that arrest continuous execution. Consequently, the bitwise complement $\neg x = x \oplus 1$ emerges as the unique fixed-point-free bijection on the Boolean domain $\{0,1\}$.

When distributed over an n-bit register $x = x_0 x_1 \ldots x_{n-1}$, the non-terminal constraint permits inversion only where a bit exits the boundary. This restriction establishes the twisted shift operator τ:

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$$\tau(x_0x_1 \dots x_{n-1}) = x_1x_2 \dots x_{n-1}\neg x_0 = \text{rotl}_1(x) \oplus 1$$

This transformation corresponds to the twisted ring counter, the Johnson counter, and Golomb's complemented cycling register (CCR) within algebraic shift-register theory. When iterated across k discrete steps ($0 \leq k \leq n$), the transformation operates as block complement feedback:

$$\tau^k(x) = \text{rotl}_k(x) \oplus (2^k - 1)$$

At step n, the operator achieves uniform bitwise inversion, $\tau^n = C$, where $C(x) = x \oplus (2^n - 1)$ denotes the bitwise complement. At step $2n$, the transformation closes identically, $\tau^{2n} = I$, establishing an action of the cyclic group of order $2n$.

Extending this analysis to two-bit symbols under closed feedback $F_q(x) = x_2 \dots x_{n-1}q(x_0x_1)$ demonstrates that the two-bit transition rule is not an independent design parameter. Evaluating all $4^4 = 256$ maps of two-bit symbols reveals that exactly 24 are bijective. Filtering for derangements—maps devoid of fixed symbols—isolates 9 candidate permutations. Enforcing spatial reflection symmetry by requiring q to commute with the two-bit reversal operator J_2 eliminates 8 candidates, isolating the two-bit complement $q = C_2$ as the unique symmetric derangement. Because C_2 is identical to τ^2 evaluated over a two-bit register, the closed two-bit rule F_c is precisely the twisted loop executed at a two-bit stride:

$$F_c(x) = \text{rotl}_2(x) \oplus 3$$

At width $n = 2$, the operator τ_2 traces the Gray code cycle $00 \to 01 \to 11 \to 10 \to 00$, which constitutes a quarter-turn permutation on the discrete Boolean square.

The general placement of inverters across an n-cell ring is governed by the Inverter Parity Theorem. For any arbitrary inverter mask $w \in \{0,1\}^n$, the mapping $f_w(x) = \text{rotl}_1(x) \oplus w$ represents an affine transformation over the finite field $\mathbb{F}_2$. When the Hamming weight $\text{wt}(w)$ is even, f_w is conjugate via an affine translation to uncomplemented cyclic rotation rotl_1, which possesses two invariant attractors (0^n and 1^n) and behaves as a static latch. When $\text{wt}(w)$ is odd, f_w is conjugate to τ, preserving zero fixed points and operating as a dynamic clock. Non-terminal continuation strictly selects the odd-parity clock sheet.

The Nested Partner Ring and Involutive Symmetry Structure

The register space $\{0,1\}^n$ is organized by three primary involutions:

- Bit reversal, defined by $J(x_0x_1 \dots x_{n-1}) = x_{n-1} \dots x_1x_0$.
- Bitwise complementation, defined by $C(x) = x \oplus (2^n - 1)$.
- The partner involution P, defined piecewise:

$$P(x) = \begin{cases} J(x), & \text{if } J(x) \neq x \\ C(x), & \text{if } J(x) = x \end{cases}$$

Because palindromic words ($J(x) = x$) are mapped to their complements ($C(x)$), which are themselves palindromes ($J(C(x)) = C(J(x)) = C(x)$), and non-palindromic words map to non-palindromes, P constitutes an involution without fixed points ($P^2 = I$, $P(x) \neq x, \forall x$). Every configuration is partitioned into an equivalence pair with canonical representative $\text{canon}(x) = \min(x, P(x))$ and orientation designated by the face bit $\text{face}(x) = [x \neq \text{canon}(x)] \in \{0,1\}$.

The nested partner ring R_n structures all $N = 2^n$ configurations into a circular sequence. The lower half (indices 0 to $M - 1$, where $M = 2^{n-1}$) contains the canonical representatives arranged monotonically, with

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the largest canonical word placed at index M and its partner at $M-1$. The upper half reflects these configurations under P to enforce the global facing law:

$$R_n[N-1-i] = P(R_n[i])$$

The circular successor operator $T(R_n[i]) = R_n[(i+1) \bmod N]$ satisfies the algebraic relations:

$$PTP = T^{-1}, \quad PC = CP, \quad S = PCT^{-1}, \quad B = S^2 = T_C T^{-1}$$

where $T_C = CTC$ represents the complement-conjugated successor. The group generated by $\langle T, P, C \rangle$ yields the full symmetric group S_{2^n} for all register widths $n \geq 3$, and collapses to the dihedral group of order 8 (D_4) at $n = 2$.

Combinatorial Dynamics, Center-Trace Decoding, and Hierarchical Locks

The Orbit Law of the Complemented Register

Because $\tau^n = C$ and $\tau^{2n} = I$, the cycle decomposition of the register under τ is rigorously constrained. Any orbit length L must divide $2n$, but L cannot divide n because $\tau^n(x) = C(x) \neq x$ for all x. Consequently, every orbit length satisfies $L = 2p$, where p divides n and the ratio $d = n/p$ is an odd integer.

A configuration x resides on an orbit of length dividing $2p$ if and only if $\tau^p(x) = C(x)$. The exact population of words generating orbits of fundamental period $2p$ is obtained through Möbius inversion across the odd divisors of p:

$$G(p) = \sum_{\substack{e \mid p \\ e \text{ odd}}} \mu(e) 2^{p/e}$$

The total number of disjoint closed orbits across the 2^n state space evaluates to Golomb's complemented cycling register formula:

$$N_{\text{orbits}}(n) = \frac{1}{2n} \sum_{\substack{d \mid n \\ d \text{ odd}}} \phi(d) 2^{n/d}$$

This combinatorial identity matches OEIS sequence A000016. Whenever n is an exact power of two ($n = 2^k$), the register admits no non-trivial odd divisors; every state belongs to an orbit of maximal length 2^{k+1}. Conversely, when n contains an odd prime factor $q > 1$, sub-orbits of shorter periods emerge.

Register Width (n)	Admissible Orbit Lengths (2p)	Word Population per Orbit Length	Total Orbit Count (OEIS A000016)
2	4	4	1
4	8	16	2
6	4, 12	4 (at $L = 4$), 60 (at $L = 12$)	6

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Register Width (n)	Admissible Orbit Lengths (2p)	Word Population per Orbit Length	Total Orbit Count (OEIS A000016)
8	16	256	16
10	4, 20	4 (at $L = 4$), 1,020 (at $L = 20$)	52
12	8, 24	16 (at $L = 8$), 4,080 (at $L = 24$)	172
64	128	2^{64}	2^{57}
192	128, 384	2^{64} (at $L = 128$), $2^{192} - 2^{64}$ (at $L = 384$)	$(2^{192} + 2 \cdot 2^{64})/384$

The Center Reader and Temporal Reconstruction

The native inward routing mechanism evaluates concentric mirror pairs $g_j(x) = x_j \oplus x_{n-1-j}$ sequentially from the boundary inward, terminating at the innermost mirror difference:

$$r(x) = x_{h-1} \oplus x_{n-h}, \quad h = \lfloor n/2 \rfloor$$

The scalar bit $r(x)$ serves as the terminal decision root of the routing tree. Under the dynamic evolution generated by τ, the center reading at tick t, denoted $y_t = r(\tau^t x)$, sequentially scans adjacent circular edge differences $D_j = x_j \oplus x_{(j+1) \bmod n}$.

For an even register width $n = 2h$, the temporal trace is governed by an exact correspondence:

$$y_t = D_{(h-1+t) \bmod n} \oplus [t = h]$$

The discrete indicator $[t = h]$ represents a deterministic phase inversion generated when the twisted register boundary crosses the center observation aperture. This produces three mathematical properties:

First, the circular summation of adjacent differences across any closed ring vanishes identically ($\bigoplus_{j=0}^{n-1} D_j = 0$). Because exactly one temporal reading is inverted by the twisted boundary, the n sequential center readings sum to an invariant odd parity:

$$\bigoplus_{t=0}^{n-1} y_t = 1$$

Second, the initial $n - 1$ center readings $y_0, \dots, y_{n-2}$ completely determine all circular edge differences D_j. Integrating these differences recovers the complete word configuration up to global complementation. The native face bit face(x) uniquely resolves the complement gauge, permitting full reconstruction of the n-bit word in linear time without state search.

Third, under the two-bit clock $F_c = \tau^2$, sampling exclusively at even ticks ($t = 0,2, \dots, n-2$) observes a single perfect matching of h edges across the cycle graph C_n, yielding rank $h = n/2$. The interleaved odd-

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phase sampling supplies the complementary matching; combined, the dual-phase readings attain rank $n - 1$.

Odd-Lane Compiler Locks (Λ_q)

When a register width factors as $n = q \cdot s$, the sub-orbits satisfying $\tau^s(x) = C(x)$ correspond to synchronized compiler sub-units. These invariant configurations exist if and only if q is odd. The resulting states form alternating complemented lane vectors:

$$\Lambda_q(u) = u \parallel \bar{u} \parallel u \parallel \bar{u} \parallel \cdots \parallel u$$

where u is an s-bit word, $\bar{u} = C(u)$, and the full register comprises q alternating blocks. If q is even, the condition $\tau^s(x) = C(x)$ yields zero solutions because shifting an even number of alternating blocks forces a circular boundary mismatch ($u = \bar{u}$).

The injection operator Λ_q serves as a compiler homomorphism, commuting with all native ring operations:

$$\tau_n \Lambda_q = \Lambda_q \tau_s, \quad J_n \Lambda_q = \Lambda_q J_s, \quad C_n \Lambda_q = \Lambda_q C_s, \quad P_n \Lambda_q = \Lambda_q P_s$$

Consequently, $\text{canon}_n(\Lambda_q u) = \Lambda_q(\text{canon}_s u)$ and $\text{face}_n(\Lambda_q u) = \text{face}_s(u)$. When s is even, the global center reading inherits the local core reading: $r_n(\Lambda_q u) = r_s(u)$.

The temporal center trace of a locked state $\Lambda_q(u)$ exhibits minimal fundamental period $p = s$. Utilizing the Knuth-Morris-Pratt (KMP) prefix-function algorithm, a 192-bit register constructed from $q = 3$ lanes of $s = 64$ bits is certified as three locked 64-bit compilers solely by verifying that its center trace has period 64. Sampling $s - 1 = 63$ temporal bits alongside the single global face bit reconstructs the core state u, which projects via Λ_3 to yield the complete 192-bit state without inspecting individual lanes.

Interface Mechanics: Defect Coordinates, Transport, and Address Geometry

Defect Closure and Skew-Product Transport

To evaluate states outside the synchronized lock families, an arbitrary state $X \in \{0,1\}^{qs}$ is decomposed into reference and defect coordinates. Defining $V_i = L_i \oplus c_i$, where L_i denotes the i-th s-bit lane and $c_i = 1^s$ for odd i (0^s for even i), the configuration is mapped bijectively to:

$$u = V_0, \quad E_i = V_i \oplus V_0 \quad (i = 1, \ldots, q-1)$$

The locked condition corresponds strictly to $E_1 = E_2 = \cdots = E_{q-1} = 0$.

The global defect syndrome is defined by:

$\Delta(X) = X \oplus C(\tau^s X)$The component lanes of $\Delta(X)$ represent the cyclic differences $d_i = V_i \oplus V_{(i+1) \bmod q}$ [cite: 1]. For odd q, these differences satisfy exact zero-sum closure [cite: 1]:$\bigoplus_{i=0}^{q-1} d_i = 0$

For even q, a persistent holonomy arises: $\bigoplus_{i=0}^{q-1} d_i = 1^s$, confirming that topological closure cannot occur for even lane counts.

Under closed feedback F_c, one clock tick transforms (u, E) via a skew product:

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$u' = F_c(u) \oplus \text{top}_2(E_1)$

$E_i' = \text{shift}_2(E_i) \mid \text{top}_2(E_{i+1} \oplus E_1), \quad 1 \le i \le q-2$Over a half-revolution of $s/2$ ticks, the defect $E_{q-1}' = \text{shift}_2(E_{q-1}) \mid \text{top}_2(E_1)$
coordinate evolves according to a linear transformation [cite: 1]:$E \mapsto M_q E = (E_2 \oplus E_1,\ E_3 \oplus E_1,\ \ldots,\ E_{q-1} \oplus E_1,\ E_1)$

The operator M_q acts over $\mathbb{F}_2$ with order exactly q. For $q = 3$, the matrix $M_3 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ satisfies $M_3^3 = I$.

Because $\Delta(F_c X) = \text{rotl}_2\Delta(X)$, closed feedback preserves the total Hamming weight of the defect syndrome: $\text{wt}(\Delta(F_c X)) = \text{wt}(\Delta(X))$. This establishes the No Closed Entry Theorem: no configuration can transition into or out of a locked family under closed feedback dynamics. Compiler locks represent dynamically isolated sub-manifolds that can only be entered or exited via open-port boundary injections or non-invertible projections.

Pulse Deviation Schedules and Error-Correction Interfaces

When an external two-bit symbol b enters via an open port F_b, it induces a deviation $\varepsilon = b \oplus C_2(a)$ relative to closed feedback, where $a = x_0 x_1$ represents the departing two-bit symbol. The deviation propagates through the register as a localized two-bit pulse:

$$X_t \oplus X_t^{\text{undeviated}} = \text{rotl}_2^{t-1}(\varepsilon)$$

The perturbation moves two bit positions per clock cycle toward the central reader and the reference lane. The arrival times at the central detection window depend on the parity of $h = n/2$:

- When h is even, the deviation $\varepsilon = 10_2$ arrives at tick $t \equiv h/2 \pmod h$; $\varepsilon = 01_2$ arrives at $t \equiv h/2 + 1 \pmod h$; and $\varepsilon = 11_2$ registers across both ticks.
- When h is odd, both $\varepsilon = 01_2$ and $\varepsilon = 10_2$ register at $t \equiv (h+1)/2 \pmod h$, whereas $\varepsilon = 11_2$ flips both central bits simultaneously, leaving the center difference $r(x) = x_{h-1} \oplus x_h$ unchanged and rendering the double pulse invisible at the center aperture.
- The perturbation reaches the reference lane L_0 at tick $t_{\text{ref}} = (q-1)s/2 + 1$. For all odd $q \ge 3$, $t_{\text{ref}} > t_{\text{center}}$, establishing that the center registers boundary deviations before they influence the reference compiler core.

Lane Width (s)	Lane Count (q)	Total Bits (n)	Pulse ε=10 Arrival	Pulse ε=01 Arrival	Pulse ε=11 Arrival	Reference Lane Arrival (tref)
64	3	192	$48 + 96k$	$49 + 96k$	Both (48,49)	65
64	5	320	$80 + 160k$	$81 + 160k$	Both (80,81)	129

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Lane Width (s)	Lane Count (q)	Total Bits (n)	Pulse ε=10 Arrival	Pulse ε=01 Arrival	Pulse ε=11 Arrival	Reference Lane Arrival (tref)
64	7	448	$112 + 224k$	$113 + 224k$	Both (112,113)	193
62	3	186	$47 + 93k$	$47 + 93k$	Never detected	63
62	5	310	$78 + 155k$	$78 + 155k$	Never detected	125

When managing faults within a lock, de-staggering the lanes reveals that each column across the q lanes forms a classical repetition code. Bitwise majority voting across columns tolerates up to $\lfloor (q-1)/2 \rfloor$ corrupted lanes. However, a subtle interaction occurs at the palindromic boundary. If a locked word u is a palindrome, a single bit flip breaks palindromic symmetry, causing the partner operator P to switch from complementation (C) to reversal (J). Applying P directly to an uncorrected corrupted state yields $n-1$ errors; hence, majority decoding must precede the application of the partner involution.

Discrete Address Geometry

The location of a canonical word $c = A \cdot 2^h + B$ on the ring R_n is described analytically by the discrete parabolic rank formula:

$$\text{rank}(c) = c - \frac{c^2}{2N} + \left(\pi_c(c) - \frac{\pi(c)}{2}\right) + e$$

where $\pi(c)$ counts total palindromes strictly below c, $\pi_c(c) = \min(\pi(c), 2^{h-1})$ represents the canonical subset, and the error remainder e satisfies:

$$-h - \frac{1}{2} < e < \frac{h}{2} + \frac{1}{2}$$

The continuous quadratic profile $c - c^2/(2N)$ reflects the triangular elimination of states forced by canonical pairing. The discrete correction $\pi_c(c) - \pi(c)/2$ forms a symmetric tent function peaking at 2^{h-2} slots at the equator $c = N/2$. Empirical evaluations from $n = 4$ to 16 demonstrate that e occupies a tighter band, $-h/3 - 0.06 < e \leq 0$, whose structural origin traces to the discrepancy of two-dimensional van der Corput bit-reversal point sets.

Unitary Quantum Dynamics, Trotterization, and Hamiltonian Simulation

Quantum Operator Embedding and the 2-Qubit HP Lattice Fold

The connection between classical bijective routing and quantum mechanics is established via the Koopman-von Neumann representation: any reversible classical bijection $F: X \to X$ lifts to a unitary permutation matrix acting across an orthonormal Hilbert basis:

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$$U_F = \sum_{x \in X} |F(x)\rangle\langle x|, \quad U_F^\dagger U_F = I$$

To assess whether this ring structure compiles physical molecular folding, a coarse-grained Hydrophobic-Polar (HP) square-lattice model is constructed over a 2-qubit conformation space.

Consider the four-residue peptide sequence $H - P - P - H$, where non-bonded hydrophobic (H) contacts provide energetic stabilization. Fixing the initial bond $(0,0) \to (1,0)$ pointing east, two consecutive relative turns encode the backbone conformation, where 0 denotes a left turn and 1 denotes a right turn:

Turn Bits (bob1)	**Relative Moves**	**Cartesian Coordinates of Residues 0 to 3**	**Contact Energy (V)**	**Ring Address Basis Index**
00	Left, Left	$(0,0), (1,0), (1,1), (0,1)$	-1 (H–H contact)	$\|00\rangle$ (index 0)
01	Left, Right	$(0,0), (1,0), (1,1), (2,1)$	0 (open chain)	$\|10\rangle$ (index 1)
10	Right, Left	$(0,0), (1,0), (1,-1), (2,-1)$	0 (open chain)	$\|01\rangle$ (index 2)
11	Right, Right	$(0,0), (1,0), (1,-1), (0,-1)$	-1 (H–H contact)	$\|11\rangle$ (index 3)

In the standard turn basis $|b_0 b_1\rangle$, the contact potential operator is diagonal: $V_{\text{word}} = -\frac{1}{2}(I + Z_0 Z_1)$The width-2 ring ordering arranges states as $(|00\rangle, |10\rangle, |01\rangle, |11\rangle)$. The unitary transformation Q maps the configuration basis into ring-address coordinates via $H_{\text{ring}} = Q^\dagger H_{\text{word}} Q$. Under this reordering, the diagonal potential becomes:

$$V = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Kinetic Generators and Split-Operator Compilation

To simulate transitions between conformational states, a kinetic hopping operator is introduced. Defining T as the forward successor on the 4-state ring: $T = (0 \quad 0 \quad 0 \quad 1\,1 \quad 0 \quad 0 \quad 0\,0 \quad 1 \quad 0 \quad 0\,0 \quad 0 \quad 1 \quad 0)$A Hermitian kinetic generator modeling uniform nearest-neighbor hopping is specified as $K = -\gamma(T + T^\dagger)$. For coupling parameter $\gamma = 0.7$, the composite Hamiltonian $H = V + K$ evaluates to:

$$H = \begin{pmatrix} -1.0 & -0.7 & 0.0 & -0.7 \\ -0.7 & 0.0 & -0.7 & 0.0 \\ 0.0 & -0.7 & 0.0 & -0.7 \\ -0.7 & 0.0 & -0.7 & -1.0 \end{pmatrix}$$

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Because K is circulant, it is diagonalized by the 4-point Quantum Fourier Transform (QFT):

$$F_{kj} = \frac{1}{2} e^{2\pi i k\ /4}, \quad FKF^{\dagger} = \text{diag}(\lambda_0, \lambda_1, \lambda_2, \lambda_3)$$

with kinetic dispersion eigenvalues $\lambda_k = -2\gamma\cos(2\pi k/4)$, yielding $\vec{\lambda} = (-1.4, 0.0, 1.4, 0.0)$.

The quantum evolution operator $U(t) = e^{-iHt}$ is compiled using second-order Strang splitting across discrete time intervals $\Delta t = t/M_{\text{steps}}$:

$$S(\Delta t) = e^{-iV\Delta t/2}\, F^{\dagger}\text{diag}\left(e^{-i\lambda_k \Delta t}\right) F\, e^{-iV\Delta\ /2}$$

The circuit execution pipeline applies:

1. State preparation by Hadamard gates initializing the uniform configuration superposition $|\psi_0\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$.
2. Potential phase rotation via $\text{diag}(e^{i\Delta t/2}, 1, 1, e^{i\Delta t/2})$.
3. Forward discrete QFT basis transformation.
4. Diagonal kinetic phase propagation via $\text{diag}(e^{1.4i\Delta t}, 1, e^{-1.4i\Delta t}, 1)$.
5. Inverse QFT ($F^{\dagger}$) returning to the spatial configuration basis.
6. Terminal potential phase rotation completing the symmetric step.

Direct matrix exponentiation at dimensionless time $t = 1.0$ yields the state vector:

$$|\psi(1)\rangle \approx \begin{pmatrix} -0.374584 + 0.495541i \\ 0.036049 + 0.335896i \\ 0.036049 + 0.335896i \\ -0.374584 + 0.495541i \end{pmatrix}$$

The aggregate probability of observing a folded contact configuration ($|00\rangle$ or $|11\rangle$) increases from an initial 0.50 to 0.771749. Numerical evaluation confirms quadratic error convergence consistent with second-order splitting:

Strang Split Steps (Msteps)	**Operator-Norm Error ‖SM−e−iH‖**	**Contact State Probability (Pcontact)**
1	8.10577×10^{-2}	0.736225
4	4.84226×10^{-3}	0.769902
16	3.01747×10^{-4}	0.771636
64	1.88557×10^{-5}	0.771742

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Strang Split Steps (Msteps)	Operator-Norm Error ‖SM−e−iH‖	Contact State Probability (Pcontact)
Direct Exponentiation (∞)	Reference (0.0000)	0.771749

Cryptographic and Biological Query Execution

Query Routing Architecture: Contextual Parsing vs. Relational Routing

The interface design enforces a strict separation of computational responsibilities: conventional neural language models handle natural-language context, identify semantic intent, and format rigorously bounded relational queries. The ring system operates as a deterministic logic router, resolving exact bit strings, hashes, and sequence constraints without speculative inference. The conventional AI then translates the discrete output back into contextual natural language.

This structure requires distinguishing query identity from relational resolution. A cryptographic hash or DNA sequence can serve as an exact identifier or an operational query target. Searching for an associated record, verifying a proposed candidate, and attempting non-brute-force preimage recovery constitute fundamentally different operations. The logic router halts if a query lacks necessary constraints, preventing generative models from introducing speculative qualifiers.

SHA-256 Preimage Verification vs. Inversion Complexity

The framework has been applied to analyze cryptographic hash functions, specifically evaluating backward-walk execution across SHA-256 round operations. Because each round transformation in SHA-256 is algebraically bijective when the schedule word W_i and constant K_i are provided, an exact backward round function exists:

$$\begin{aligned}
&\text{old_a} = b_n, \quad \text{old_b} = c_n, \quad \text{old_c} = d_n \\
&\text{old_e} = f_n, \quad \text{old_f} = g_n, \quad \text{old_g} = h_n \\
&T_2 = \Sigma_0(\text{old_a}) + \text{Maj}(\text{old_a},\text{old_b},\text{old_c}) \\
&T_1 = a_n - T_2 \\
&\text{old_d} = e_n - T_1 \\
&\text{old_h} = T_1 - \Sigma_1(\text{old_e}) - \text{Ch}(\text{old_e},\text{old_f},\text{old_g}) - K_i - W_i
\end{aligned}$$

Tests executed in compiled C demonstrated recovery of 4-byte preimages (e.g., !ABC) from hash digests in 161.7 seconds across 5.57×10^8 backward evaluations (a processing rate of 3.5×10^6 walks/second).

However, the theoretical audit clarifies the cryptographic boundaries of this mechanism:

1. The backward walk functions as an exact verification oracle, checking all 256 state bits against the initial initialization vector (H_0) simultaneously without false positives.
2. Because the message schedule word W_0 remains unknown, the backward walk must iterate over candidate inputs, meaning the search space scales as $2^{8n_{\text{bytes}}}$. For a 4-byte message, the search space is $2^{32} \approx 4.29 \times 10^9$; for 8 bytes, it reaches 2^{64}, rendering exhaustive backwards verification computationally intractable.
3. Translating the 64 rounds of SHA-256 into 328 algebraic equations within the Z3 theorem prover resolved a single-byte preimage ('A') in 54.74 seconds via constraint propagation without iterative

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guessing. However, it timed out on other single-byte inputs ('N') because Boolean satisfiability (SAT) solvers encounter exponential DPLL branching across non-linear modulo-addition carry chains.

The carry bits generated during modulo additions (Δ-channel exhaust) represent the structural cost of non-linear folds. Unless an explicit mathematical relation derives the carry bit profile directly from the digest without candidate enumeration, the one-way property of SHA-256 remains an intact complexity-theoretic barrier.

Comprehensive Evaluation: Verified Mechanisms vs. Missing and Open Links

To delineate the verified mechanics of the system from its unresolved claims, the following matrix compiles all components established across the One Inverter corpus and quantum compiler tests:

Theorem / Component	Mathematical Description	Formal Verification Scope	Status	Source
Theorem 3.1	One-bit fixed-point freedom forces NOT	Exhaustive (all 4 maps of $\{0,1\}$)	Proved	One Inverter §3
Theorem 3.2	$\tau^k = \text{rotl}_k \oplus (2^k - 1)$, $\tau^n = C$, $\tau^{2n} = I$	Exhaustive, $n = 2 \ldots 16$, all k	Proved	One Inverter §3
Theorem 3.3	Two-bit derangement filtration yields $C_2 = P_2 = \tau^2$	Exhaustive across all 256 two-bit rules	Proved	One Inverter §3
Proposition 3.4	Width-2 τ is Gray quarter-turn; T_2 breaks square edges	Exhaustive on S_4 permutations	Proved	One Inverter §3
Theorem 3.5	Inverter parity: even mask → latch (2 fixed), odd → clock (0 fixed)	Exhaustive $n = 6 \ldots 10$; sampled $n = 12$	Proved	One Inverter §3
Theorem 4.1	Golomb orbit spectrum: lengths $2n/d$ for odd $d \mid n$; population $G(p)$	Exhaustive, $n = 2 \ldots 20$ (2,097,148 states)	Proved	One Inverter §4
Theorem 4.2	$\tau^s(x) = C(x) \Leftrightarrow x = \Lambda_q(u)$ with q odd	Exhaustive, all $q \cdot s \leq 18$	Proved	One Inverter §4

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Theorem / Component	Mathematical Description	Formal Verification Scope	Status	Source
Theorem 5.1	Routing root $r(x) = x_{h-1} \oplus x_{n-h} = g_{h-1}(x)$ (innermost lamp)	$n = 2 \ldots 14$ exhaustive; 1,800 wide states	Proved	One Inverter §5
Theorem 5.4	Temporal reader: $y_t = D_{(h-1+t) \bmod n} \oplus [t = h]$; reconstruction	Exhaustive, even $n = 2 \ldots 16$	Proved	One Inverter §5
Theorem 6.1	Reading poset ≅ ∂(cross-polytope) ≅ S^{n-1}; quotient $\mathbb{R}P^{n-1}$	Homology/Betti numbers, $n = 1 \ldots 7$	Proved	One Inverter §6
Proposition 6.3	Adjacency: P tears $n2^{n/2}$ edges (even n), jumping dist $n-1$	Exhaustive edge evaluation, $n = 3 \ldots 14$	Proved	One Inverter §6
Theorem 7.1	Homomorphism: Λ_q intertwines τ, J, C, P; multiplies d_H by q	6,132 lifts exhaustive; 200 wide tests	Proved	One Inverter §7
Theorem 7.3	Trace period p certifies lock $x = \Lambda_{n/p}(u)$ in linear time	$n = 2 \ldots 16$ exhaustive; up to 1,920 bits	Proved	One Inverter §7
Theorem 8.1	Defect bijection (u, E); closure $\oplus d_i = 0$ (odd q)	1,332,288 states exhaustive; $q = 2 \ldots 8$	Proved	One Inverter §8
Theorem 8.2	Transport: M_q acts with exact order q	Exhaustive; $q = 3 \ldots 15$	Proved	One Inverter §8
Corollary 8.4	Invariance: wt($\Delta(X)$) conserved; no closed entry to locks	Analytic proof from affine linearity	Proved	One Inverter §8

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Theorem / Component	Mathematical Description	Formal Verification Scope	Status	Source
Theorem 8.5	Deviation arrival times at center and lane o for even/odd h	297 injected pulses, $s \leq 130, q \in \{3,5,7\}$	Proved	One Inverter §8
Theorem 8.9	Rank parabola: rank$(c) = c - c^2/2N +$ tent $+ e$, bounds on e	Exhaustive canonical words, $n = 4 \ldots 16$	Proved	One Inverter §8
Quantum Model	2-qubit HP lattice Strang Trotter circuit; QFT diagonalization	Unitarity residual $< 10^{-15}$; 2nd order error	Verified	Quantum Fold Bundle
Preimage Walk	Invertible backward round execution at 3.5×10^6 states/sec	4-byte message recovery in 161.7 sec	Verified	C-Tbone Source

Foundational Missing Links

A critical assessment reveals five foundational gaps that separate the proven finite-combinatorial mechanics from the broader claims regarding quantum compilation and biological protein folding:

First, the physical origin of net register inversion remains unidentified. Theorem 3.5 proves that an odd number of inverters forces an un-halting dynamic clock, whereas an even number creates a static latch. However, the physical mechanism that enforces an odd inversion in macromolecules (such as polypeptide chains or nucleic acid transcription complexes) is unestablished. The framework operates on an abstract sign rule ($\Phi = \pi$) without deriving which physical Hamiltonian term, steric constraint, or geometric phase supplies this topological twist.

Second, molecular Hamiltonians have not been derived from ring coordinates. The 2-qubit HP lattice simulation confirms that a contact energy potential V can be transformed into ring coordinates via $Q^\dagger V Q$, and that cyclic hopping $K = -\gamma(T + T^\dagger)$ can be compiled via a Fourier split-operator circuit. However, this transition generator K represents an asserted ring-adjacency model rather than an interaction derived from molecular mechanics. A continuous single-turn flip mixer transformed into ring coordinates produces a Hamiltonian distinct from ring-neighbor hopping. Furthermore, the simulation describes a closed, unitary system exhibiting undamped Rabi oscillations (P_{contact} oscillates between 0.50 and 0.77). Real biological folding is an open, dissipative thermodynamic process governed by master equations or Langevin dynamics that drive the peptide down an energetic funnel toward an equilibrium native fold. The toy model also excludes 3D Ramachandran dihedral angles, side-chain rotamers, solvent entropy, and chaperone ATP consumption.

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Third, rules for open-port entry and switching between locked compiler families remain open. Corollary 8.4 proves that under closed feedback, a compiler lock cannot be entered or exited because $\mathrm{wt}(\Delta(X))$ is strictly conserved. Locked compilers therefore form isolated sub-manifolds. The grammar governing how open-port deviations or external environment signals switch states between distinct odd-lane lock families without compromising reversibility has not been formulated.

Fourth, asymptotic mathematical conjectures remain unproved. The proposition that $B = T_C T^{-1}$ has no fixed points is verified computationally up to $n = 22$, but an analytical proof across all widths n requires demonstrating that the address complement map $\kappa(i)$ can never satisfy $\kappa(i) = \kappa(i-1) + 1$. Similarly, the rank remainder e measures empirically within $[-h/3 - 0.06, 0]$, well inside the proved band $(-h - 1/2, h/2 + 1/2)$. An analytical derivation connecting this tighter bound to the discrepancy of two-dimensional bit-reversal point sets remains incomplete. In addition, while cycle decompositions of $S = PCT^{-1}$ display consistent empirical structures (such as the $27 \to 28$ branch transition between $n = 8$ and $n = 10$), a universal algebraic phase rule across widths has not been established.

Fifth, relational logic routing must be distinguished from non-brute-force preimage recovery. The logic router deterministically resolves relations given sufficient constraints. However, establishing that individual SHA-256 round operations are invertible does not bypass the exponential complexity of modulo-addition carry propagation when input words are unknown. The Δ-channel exhaust (carry bits) tracks where non-linear folding occurs, but extracting the preimage directly from the output digest without candidate enumeration remains bounded by the complexity of NP-hard Boolean satisfiability.

Algorithmic Audit of the Verification Testbed

The computational testbed NEXUS_143_one_inverter_verification.ipynb operates self-contained verification code using the Python standard library and NumPy, executing 51 distinct assertions across 10 functional groups. The execution integrity is anchored by an initial SHA-256 seal:

SEAL = 009815944fd8951cf95ca6bd56728f54b5e20634dd03e2d8cc67e005019d1dea

The test runner terminates with an assertion failure if the hash of the prediction text deviates from this digest before any computational routine executes.

The numerical audit confirms the execution parameters of the 51 verification checks:

- In Group 2, $PTP = T^{-1}$, $PC = CP$, and $S^2 = T_C T^{-1}$ were verified across all words for $n = 2 \ldots 22$. The operator $B = T_C T^{-1}$ was confirmed to have zero fixed points across all $4{,}194{,}304$ words at $n = 22$ ($\langle T, T_C \rangle_F = 0$). The group $\langle T, P, C \rangle$ was verified to equal S_{2^n} for $n = 3 \ldots 8$ via 2-transitivity and transposition generation.
- In Group 3, NOT was verified as the unique fixed-point-free unary Boolean map. The relation $\tau^k(x) = \mathrm{rotl}_k(x) \oplus (2^k - 1)$ was validated for all $x \in \{0,1\}^n, k \le n, n = 2 \ldots 16$. Two-bit filtration counts ($256 \to 24 \to 9 \to 1$) and inverter mask conjugacy were validated across 2,112 masks up to $n = 12$.
- In Group 4, cycle decompositions for $2{,}097{,}148$ words across $n = 2 \ldots 20$ matched OEIS A000016. The identity $\tau^s(x) = C(x)$ was confirmed to yield exactly 2^s solutions for odd q and zero solutions for even q across all $q \cdot s \le 18$.
- In Group 5, the routing root $r(x)$ was verified to equal the innermost lamp $g_{h-1}(x)$ across all words for $n = 2 \ldots 14$ and across 1,800 random samples at widths $63, 64, 127, 128, 192, 256$. The six baseline

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text inputs (Ovaltine and Singularity strings at 64, 128, and 256 bits) were verified against predicted routing outputs. Temporal reader parity ($\bigoplus y_t = 1$), decoding from $n-1$ readings plus the face bit, and rank distributions were validated.

- In Group 6, face complexes of the cross-polytope boundary $\partial\Diamond^n$ yielded reduced $\mathbb{Z}_2$-Betti numbers $\tilde{\beta}_{n-1} = 1$ (sphere S^{n-1}) and uniform Betti numbers equal to 1 for the projective quotient $\mathbb{R}P^{n-1}$ for $n = 2 \ldots 7$. Edge-tearing counts for P matched $n2^{n/2}$ at even n and $(n-1)2^{(n+1)/2}$ at odd n, with all torn edges mapping to Hamming distance $n-1$.
- In Group 7, Λ_q homomorphism relations were verified across 6,132 lifts ($s \leq 10, q \in \{3,5,7\}$) and 200 wide tests ($s \in \{64,128\}$). Trace period spectra confirmed that 192-bit states partition into 2^{64} states of period 64 and $2^{192} - 2^{64}$ states of period 192. For $u =$ 0x446f6e7420666f72, the 192-bit lift $\Lambda_3(u)$ was confirmed to have trace period 64, with its initial 63 center bits and face bit 0 returning u, and its ring address matching 0x3b49bcdccb9d2d2b3aa42831cb4f5581dfa12a793ffbdf03.
- In Group 8, defect coordinate bijections were validated across $1{,}332{,}288$ states. The operator M_q was confirmed to have order q across odd $q = 3 \ldots 15$. Pulse arrival schedules were verified across 297 injected pulses, and address parabola bounds were validated across canonical words for $n = 4 \ldots 16$.

All 51 verification checks passed without failure.

Synthesis and Strategic Outlook

The examined framework provides an exact, finite-combinatorial foundation for reversible register routing, temporal center-trace decoding, and hierarchical sub-compiler locking. Its mathematical contributions include:

- Demonstrating that the principle of non-terminal continuation selects twisted complement feedback (τ) as the unique dynamic clock.
- Proving that an n-bit configuration can be reconstructed from $n-1$ scalar sequential center measurements combined with a single geometric orientation bit.
- Formulating compiler nesting through odd-lane alternating complement embeddings ($\Lambda_q(u)$), which preserve the full automorphism group of the small compiler within the larger register.
- Establishing defect transport mechanics and the No Closed Entry Theorem, proving that defect syndromes rotate without decay under closed feedback.

However, connecting this discrete register logic to quantum compilation and molecular biology requires resolving fundamental theoretical gaps:

1. Deriving molecular Hamiltonians from first principles: replacing arbitrary ring-hopping generators ($K = -\gamma(T + T^{\dagger})$) with physical interaction potentials derived from polypeptide dihedral kinematics and non-bonded force fields.
2. Incorporating open thermodynamic environments: replacing closed unitary Rabi oscillations with master equations that describe the dissipative, non-equilibrium funnel kinetics necessary to resolve Levinthal's paradox.

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3. Identifying the physical inversion mechanism: determining the specific steric, topological, or enzymatic interaction that enforces net odd register inversion ($\Phi = \pi$) in macromolecular systems.
4. Resolving open mathematical conjectures: developing analytical proofs for the fixed-point freedom of $B = T_C T^{-1}$ for all $n > 22$, classifying universal S-cycle branch templates, and establishing the exact discrepancy constant governing the rank error remainder e.

The system demonstrates that structured binary registers possess intrinsic geometric and algebraic invariants that enable exact logic routing and unitary circuit compilation. Establishing whether this architecture describes the physical mechanism of protein folding depends on replacing formal mathematical analogies with explicit mappings between register operations and biochemical interactions.

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